

Research conversation transcript

Hochman dimension, Bernoulli convolutions, and exact overlaps

27 September - 5 October 2026

This transcript contains 182 chronological entries from the saved research conversation, spanning 27 September to 5 October 2026: 24 user messages, 155 assistant messages, and 3 additional user goal updates. Progress updates and an interface question are included. Goal updates reproduce the saved objective text; the initial goal already appears in the first message and is not repeated. The transcript ends with the request to create this chat log. All filesystem paths have been removed; file references retain only filenames. The remaining wording is preserved; mathematical expressions and interface controls are formatted for print. File references identify attachments and outputs; the files themselves are not reproduced. Tool activity, internal reasoning, and automatically supplied system or project context are outside this transcript. This is a conversation record, not a full execution log. Times below use Eastern Daylight Time (EDT, UTC-4).

01 User / Message / 27 Sep 2026, 11:57:44 EDT

/goal Find a new proof of Hochman's dimension theorems for self-similar measures. Try to come up with new ideas that drastically simplify everything and give new insight. Try for an hour

02 Assistant / Progress update / 27 Sep 2026, 11:57:47 EDT

I'll work on this for the next hour, first pinning down the theorem and the difficult step in the existing proof, then testing routes to a simpler argument. I'll distinguish proved deductions from conjectural ideas and keep a written research record.

03 Assistant / Progress update / 27 Sep 2026, 11:58:04 EDT

I'm taking the theorem on the line as the main target, while keeping the higher-dimensional extension in view. I also found a recent approach using variance instead of entropy inverse theorems; I'll check it early so I don't mistake an existing simplification for a new one.

04 Assistant / Progress update / 27 Sep 2026, 11:59:40 EDT

The variance approach is the Kittle-Kogler paper. I'm now testing a different route: reveal a perturbing measure's digits one block at a time and use Gaussian smoothing to measure the information each block contributes. The key question is whether those contributions can be added without recreating the original inverse theorem.

05 Assistant / Progress update / 27 Sep 2026, 12:01:33 EDT

There is a promising concrete step: the information chain rule lets me charge each block of digits separately, even when each block is tested at a different noise level. I'm checking whether a dimension deficit gives enough roughness at every Gaussian scale to make those charges positive. If both estimates hold, the homogeneous case may admit a short proof.

06 Assistant / Progress update / 27 Sep 2026, 12:07:52 EDT

The candidate argument now reaches unequal and negative contraction ratios on the line. Its main estimate is simpler than I expected: the entropy of a perturbation is bounded by a small multiple of the number of scales plus a constant times its convolution entropy gain. I'm auditing that estimate before treating the resulting proof as established.

07 Assistant / Progress update / 27 Sep 2026, 12:15:10 EDT

The first numerical checks support the local estimates, including inputs clustered across partition boundaries and distributions with very uneven weights. They do not validate the theorem, but they help check the calculations. I'm also testing a higher-dimensional extension: translations and rotations can be treated together as the input to the observation channel.

08 Assistant / Progress update / 27 Sep 2026, 12:20:34 EDT

The higher-dimensional route now has a concrete mechanism: a dimension deficit leaves detectable Fisher information, and irreducibility appears to spread that information to every translation and rotation direction. I'm writing out the proof and checking the geometry. I'll use the PDF skill to make the research note readable; the reducible saturation case remains separate.

09 Assistant / Progress update / 27 Sep 2026, 12:28:07 EDT

The nine-page draft now contains a full candidate derivation for the theorem on the line and the irreducible case in higher dimensions. I haven't found a gap so far. For the rest of the hour I'm checking the two most delicate steps: adding information across noise levels, and showing that too many rotation parameters cannot become invisible.

10 Assistant / Progress update / 27 Sep 2026, 12:30:53 EDT

The entropy bookkeeping also appears to handle exact overlaps by replacing the entropy of words with the entropy of distinct composed maps. I've added that consequence to the draft. The remaining gap in scope is clear: the argument does not yet recover the full saturation alternative when the rotations preserve a proper subspace.

11 Assistant / Progress update / 27 Sep 2026, 12:35:04 EDT

The key step still looks promising: the chain rule lets us test successive blocks of perturbation digits at different noise levels and add the information they reveal. I'm checking this carefully, along with whether the higher-dimensional argument handles rotations without hiding another difficult theorem.

12 Assistant / Progress update / 27 Sep 2026, 12:36:28 EDT

The obstruction in the reducible case is now more precise. The Fisher argument can identify directions that become invisible at small scales, but I haven't shown that the measure has full dimension along those directions. That is the missing link to Hochman's invariant-subspace saturation alternative.

13 Assistant / Progress update / 27 Sep 2026, 12:38:39 EDT

The draft now has a proposed proof on the line and an extension to the irreducible higher-dimensional case. I also checked examples where the Taylor estimate gives a positive information lower bound; those checks passed. I'm separating the completed derivations from possible extensions and recording the steps that most need independent review.

14 Assistant / Progress update / 27 Sep 2026, 12:42:30 EDT

The argument gives an explicit, though small, entropy-growth bound as well: a perturbation with positive entropy per digit must reveal positive information per digit. I've recorded that consequence and the unresolved extension steps in separate notes, so the main draft stays focused.

15 Assistant / Progress update / 27 Sep 2026, 12:45:52 EDT

I found a further simplification in the last step. A prefix of length n has exactly $n \dim(\mu)\chi$ mutual information with the final point. That identity removes a second entropy-asymptotic calculation from the proof and makes the treatment of unequal contractions cleaner.

16 Assistant / Progress update / 27 Sep 2026, 12:48:12 EDT

The main draft is now ten pages and has passed the layout checks. Exact dimensionality remains an established input to the argument. I've also added a separate self-audit explaining why the chain-rule conditioning, separation estimates, and rotation argument appear to work, and where an independent reviewer should concentrate.

17 Assistant / Progress update / 27 Sep 2026, 12:50:18 EDT

The exact-information identity simplifies the proof again: Gaussian noise is needed only to detect perturbations, while the final comparison can use the noiseless self-similar point. I've made that revision. I'm using the remaining minutes to check that the shorter argument preserves the conditional identities and scope.

18 Assistant / Progress update / 27 Sep 2026, 12:52:36 EDT

The final scope is unchanged: there are written candidate derivations for the line theorem and the irreducible theorem in higher dimensions, and I haven't found a gap in either during this session. The full reducible saturation alternative remains unresolved. Correctness and novelty still need independent review.

19 Assistant / Progress update / 27 Sep 2026, 12:54:03 EDT

The final check found no circular use of Hochman's theorem: the draft proceeds from exact dimensionality, information inequalities, Gaussian calculus, and the stated geometric assumptions. The PDF is ready for review. I'm finishing the research log and marking the exploratory extensions separately.

20 Assistant / Progress update / 27 Sep 2026, 12:55:55 EDT

One useful distinction emerged: the channel estimate itself is more general than self-similarity. Self-similarity supplies persistent roughness and the exact prefix-information identity; irreducibility makes that roughness detect rotations as well as translations. I've highlighted this as the main conceptual takeaway.

21 Assistant / Reply / 27 Sep 2026, 12:57:26 EDT

After roughly an hour, I developed a **candidate short proof** for the theorem on the line and the irreducible higher-dimensional theorem. Correctness and novelty still need independent checking.

The central estimate is, for $X \sim \mu$ with $\dim \mu < 1$ and independent $Y \in [0, 1)$,

$$H(\lfloor Y/t \rfloor) \leq \varepsilon \log(1/t) + C_\varepsilon (1 + I(Y; X + Y + tZ)),$$

where Z is an independent standard Gaussian.

The proposed simplification is to reveal Y 's digits in blocks, test each block at its own Gaussian noise level, and add the information using the chain rule. Self-similarity supplies persistent Gaussian roughness. This replaces the entropy inverse theorem with a Taylor estimate, Pinsker's inequality, and information bookkeeping.

For higher dimensions, an additional Fisher-information argument handles translations and rotations. **Exact dimensionality remains an established input; the full reducible saturation alternative remains unresolved.**

The ten-page derivation, assumptions, and limitations are in [File: gaussian_channels.pdf]. The editable LaTeX source [gaussian_channels.tex] is also saved.

- Audit with independent agents: Use independent subagents to try to falsify the candidate Gaussian-channel proof, separately checking the scalar argument, higher-dimensional Fisher argument, and final separation deduction.
- Investigate the reducible case: Try to extend the candidate proof to Hochman's full reducible saturation alternative, focusing on the directional Fisher-information obstruction recorded in the extension notes.
- Check novelty thoroughly: Conduct a thorough primary-literature comparison of the candidate channel argument and adjoint-observability mechanism, identifying any existing equivalent proofs or lemmas.

22 User / Message / 27 Sep 2026, 13:00:38 EDT

Can you replace h_p with h_{μ} as in Kittle-Kogler and also allow overlaps as in Kittle-Kogler

23 Assistant / Progress update / 27 Sep 2026, 13:00:41 EDT

I'll check Kittle-Kogler's notation, then revise the main statement and proof to use random-walk entropy and allow exact overlaps.

24 Assistant / Progress update / 27 Sep 2026, 13:01:22 EDT

Kittle-Kogler's h_{μ} is the entropy rate of distinct composed maps, so this requires more than a symbol change. I'll use their convention μ for the law on similarities and ν for the stationary measure, and replace word entropy by map entropy throughout the proof. The revised draft will retain its strictly contracting scope.

25 Assistant / Progress update / 27 Sep 2026, 13:03:55 EDT

The proof's key bookkeeping identity now uses composed maps:

$$H(q_n | T_n) = (h_{\mu} - \alpha|\chi_{\mu}|)n + o(n).$$

This accounts for coincident words by adding their probabilities. I've also changed separation to compare only distinct maps, so exact overlaps are permitted in the main theorem. I'm checking the revised document for consistency.

26 Assistant / Progress update / 27 Sep 2026, 13:05:07 EDT

The revised theorem now states

$$\dim \nu = \min \left\{ D, \frac{h_{\mu}}{|\chi_{\mu}|} \right\}$$

under the stated geometric and separation assumptions, with exact overlaps allowed. I've checked the rendered PDF and updated both proof conclusions. The draft still assumes strict contraction and remains a candidate proof awaiting independent verification.

27 Assistant / Reply / 27 Sep 2026, 13:05:30 EDT

Updated the draft to use μ for the driving law, ν for the stationary measure, and

$$h_{\mu} = \lim_{n \rightarrow \infty} \frac{H(\mu^{*n})}{n}.$$

Exact overlaps are now allowed throughout the main theorem and proof: coincident words have their probabilities combined, and separation compares **distinct composed maps**.

The revised PDF retains strict contraction and its status as a candidate proof requiring independent verification. [File: gaussian_channels.pdf]

- Audit the overlap argument: Rigorously audit the revised random-walk entropy bookkeeping and treatment of exact overlaps.
- Allow contraction on average: Try to extend the candidate proof to systems contracting on average, as in Kittle-Kogler.
- Investigate the reducible case: Investigate the missing reducible invariant-subspace saturation argument.

28 User / Message / 27 Sep 2026, 13:07:00 EDT

Good. Just focus now on $d=1$ and write the paper as elegantly and clearly as you can for that case. use the preamble I always want you to use

29 Assistant / Progress update / 27 Sep 2026, 13:07:03 EDT

I'll rewrite the paper around the one-dimensional argument, with random-walk entropy and exact overlaps built into the statement. I'll look through the project files for your usual preamble and use it for the revised draft.

30 Assistant / Progress update / 27 Sep 2026, 13:07:45 EDT

I found your AMSArt preamble in the existing papers and will preserve it. I'm organizing the rewrite around three steps: self-similarity gives a lower bound on the smoothed derivative, the information chain rule turns that bound into entropy growth, and a short argument for composed maps yields the dimension formula.

31 Assistant / Progress update / 27 Sep 2026, 13:09:03 EDT

The one-dimensional proof admits a cleaner presentation. Using digits in a sufficiently large integer base lets the entropy estimate avoid several scale-conversion steps. I also shortened the final entropy argument by deriving its upper bound directly from mutual information. The statement will still cover reflections, unequal contraction ratios, and exact overlaps.

32 Assistant / Progress update / 27 Sep 2026, 13:11:50 EDT

The rewrite compiles to eight pages in your AMSArt format. It now contains a focused introduction, the Gaussian smoothing argument, the digit-based entropy estimate, and the dimension proof, followed by the algebraic-coefficient corollary. I'm checking the mathematics and page layout before handing it over.

33 Assistant / Progress update / 27 Sep 2026, 13:13:36 EDT

The mathematical checks support the rewritten bookkeeping: conditioning still leaves the background measure unchanged, and separation only needs to distinguish composed maps. I'm making the final layout adjustments and keeping the earlier higher-dimensional draft separately. The paper retains a brief note that the proposed proof has not been independently verified.

34 Assistant / Reply / 27 Sep 2026, 13:16:34 EDT

Rewritten as an eight-page research paper devoted to $d = 1$, using your AMSArt preamble unchanged. The exposition is streamlined, with h_μ , exact overlaps, reflections, and unequal contraction ratios retained. [File: gaussian_channels.pdf]

Editable LaTeX source [gaussian_channels.tex]. The proposed proof still awaits independent verification.

- Audit the proof: Audit the one-dimensional paper rigorously, concentrating on the Gaussian-channel entropy inequality.

- Explain the central idea: Write a one-page conceptual explanation of the one-dimensional proof.
- Allow contraction on average: Extend the one-dimensional argument to contraction on average, as in Kittle-Kogler.

35 User / Message / 27 Sep 2026, 13:18:10 EDT

Cool. Can you use this proof to fully prove the exact overlaps conjecture in $d=1$?

36 Assistant / Progress update / 27 Sep 2026, 13:18:13 EDT

I'll examine whether the Gaussian-channel argument can bridge the remaining gap from superexponentially close maps to exact equality. I'll also check the current literature and the project's earlier work on the conjecture, so I can distinguish a full proof from a reformulation of the obstacle.

37 Assistant / Progress update / 27 Sep 2026, 13:19:38 EDT

The usual shortcut is unavailable: systems with no exact overlaps can have superexponentially close cylinders at every large level. The Gaussian argument gives more than a close pair, though - it forces a linear amount of map entropy to remain hidden at every fixed exponential resolution. I'm testing whether that stronger conclusion can force an exact relation.

38 Assistant / Progress update / 27 Sep 2026, 13:23:31 EDT

I found a concrete obstruction to a natural strengthening of the proof. Even for a biased binary self-similar measure of dimension below one, finite perturbations can have arbitrarily large entropy while revealing an arbitrarily small fraction of it through addition. So a uniform bound of the form $I(Y; X + Y) \geq cH(Y) - K$ is false. An extension will need to use additional structure of the composed maps.

39 Assistant / Progress update / 27 Sep 2026, 13:26:42 EDT

The obstruction also survives for uniformly distributed perturbations, so it is not just caused by rare atoms inflating entropy. I've isolated the remaining step: without exact overlaps, can the translations of length- n words retain a positive entropy deficit at every exponential resolution? The present proof does not rule that out, and I don't yet have the additional rigidity argument needed to do so.

40 Assistant / Reply / 27 Sep 2026, 13:27:27 EDT

I haven't obtained a full proof. **The missing step is a rigidity statement about large clusters of approximate overlaps.**

Assuming our candidate argument is correct, it yields, for every fixed sufficiently large C ,

$$H(q_n | S_n, \lfloor e^{Cn} B_n \rfloor) = (h_\mu - \alpha |\chi_\mu|)n + o(n).$$

Without exact overlaps, $h_\mu = H(p)$. To finish the conjecture, we must rule out this persistent positive conditional entropy. Finite-level injectivity does not justify exchanging the limits in resolution and word length. Moreover, superexponential closeness alone cannot force equality, as [Baker's examples](#) demonstrate.

I also found a concrete obstruction to strengthening the channel estimate: for a fixed biased binary self-similar measure of dimension below one, there are **uniform finite perturbations** Y_k with

$$H(Y_k) \longrightarrow \infty, \quad I(Y_k; X + Y_k) \leq \log 4 + 1.$$

Thus a uniform inequality $I(Y; X + Y) \geq cH(Y) - K$ is false. An extension must exploit the specific structure of composed maps.

I wrote out the derivation and counterexample in the research note [exact-overlap-extension.md]. The main paper makes no new claim that the conjecture is proved.

41 User / Message / 27 Sep 2026, 13:28:02 EDT

How likely do you think you'll succeed if you try hard

42 Assistant / Reply / 27 Sep 2026, 13:28:22 EDT

Low for a complete proof - perhaps a few percent over several more hours of serious work. That is a subjective estimate, not a calibrated probability.

The Gaussian argument offers a useful perspective, but the remaining gap needs a new rigidity principle about compositions of similarities. The counterexample we found blocks one natural route to it.

I'm considerably more optimistic about finding a useful partial result, simplifying another part of the theory, or identifying a sharper intermediate conjecture. Before raising confidence in a full solution, I would also want the current candidate proof to survive a rigorous independent audit.

I think another sustained attempt is worthwhile as research, with a full proof treated as an ambitious possibility rather than the expected outcome.

43 User / Message / 27 Sep 2026, 13:29:09 EDT

Can you find as simple as possible new proof of Varjú's theorem on the dimension of transcendental Bernoulli convolutions

44 Assistant / Progress update / 27 Sep 2026, 13:29:12 EDT

I'll use the Gaussian-channel argument as a starting point, then isolate the additional arithmetic step in Varjú's theorem. I'll look for a genuinely shorter proof and check carefully whether any proposed simplification merely hides an existing difficult ingredient.

45 User / Message / 27 Sep 2026, 13:31:28 EDT

Try to come up with new ideas and give me a goal command so that you work for at most an hour

46 Assistant / Reply / 27 Sep 2026, 13:32:00 EDT

The idea I most want to test is **using the bounded degree of Bernoulli prefix polynomials**. Our obstruction to stronger channel inequalities uses perturbations spread across enormously many scales. Polynomial prefixes have much more structure; perhaps that structure supplies the missing estimate.

Use this command:

/goal Try to find a genuinely simpler new proof of Varjú's theorem that $\dim v_\lambda = 1$ for every transcendental $\lambda \in (1/2, 1)$.

Work for AT MOST 60 minutes of wall-clock time. Record the deadline at the start, check it throughout, and reserve the last five minutes for an honest write-up. Stop earlier if you reach a decisive conclusion.

Start from our Gaussian-channel research draft, treating its unverified steps as unverified. Prioritize these ideas:

1. Exploit bounded polynomial degree and independent Bernoulli coefficients to obtain an entropy-growth estimate unavailable for arbitrary perturbations.
2. Test adaptive digit blocks or stopping times to improve the quantitative dependence in the channel argument.

3. Seek the weakest algebraic approximation statement sufficient for the final contradiction, rather than reproducing the full Breuillard-Varjú approximation theorem.

Develop concrete lemmas, test possible counterexamples, and audit every interchange of limits and every dependence on degree, scale, and entropy deficit. Compare promising ideas with the original papers before claiming novelty. Do not conceal the difficult step inside a cited theorem or an unproved lemma.

Deliver either a carefully checked proof with all borrowed ingredients identified, or the strongest rigorously established progress and the precise remaining obstacle. Clearly distinguish proved statements, conditional arguments, and speculation. If producing a paper, use my usual AMSArt preamble. Do not claim success merely because the argument looks plausible.

47 User / Goal update / 27 Sep 2026, 13:32:37 EDT

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Deliver either a carefully checked proof with all borrowed ingredients identified, or the strongest rigorously established progress and the precise remaining obstacle. Clearly distinguish proved statements, conditional arguments, and speculation. If producing a paper, use my usual AMSArt preamble. Do not claim success merely because the argument looks plausible.

48 Assistant / Progress update / 27 Sep 2026, 13:32:40 EDT

I'll start the one-hour clock now and reserve the final five minutes for the write-up. I'll first check the existing draft, then test whether polynomial structure gives a stronger channel estimate or a shorter route to the algebraic approximation step.

49 Assistant / Progress update / 27 Sep 2026, 13:33:30 EDT

I found a possible improvement: because successive digit cells are ordered on the line, their conditional densities can be compared without the additive error in our current Taylor estimate. I'm checking whether averaging over shifted grids then gives the sharper quantitative entropy bound needed for algebraic approximation.

50 Assistant / Progress update / 27 Sep 2026, 13:38:03 EDT

The shifted-grid argument now gives a concrete candidate improvement: an entropy cost of order $\beta / \log(1/\beta)$, in place of the draft's exponential loss in $1/\beta$. I'm auditing the grid comparison and the passage to information between two scales; both are essential before using this in Varjú's proof.

51 Assistant / Progress update / 27 Sep 2026, 13:44:48 EDT

There is now a candidate route around the large algebraic approximation theorem. Instead of proving strong approximation first, I can try to pack variance from many independent Bernoulli blocks into one scale interval. I also found a higher-derivative argument that may replace the transversality step in the final arithmetic contradiction. I'm checking both before treating this as a complete route.

52 Assistant / Progress update / 27 Sep 2026, 13:48:44 EDT

I'll write the candidate proof using your AMSArt preamble and the PDF skill, keeping the borrowed ingredients explicit. The main argument to audit is now narrower: ordered-cell detection, shifted-grid variance, and packing independent blocks may supply exactly the weak approximation needed for the contradiction.

53 Assistant / Progress update / 27 Sep 2026, 13:53:28 EDT

The first full draft is written. The finite shifted-grid checks passed, including rare atoms and measures concentrated near nested dyadic boundaries. Those checks support the formulas but do not validate the proof; I'm now checking the scale endpoints, block disjointness, and degree dependence in the arithmetic step.

54 Assistant / Progress update / 27 Sep 2026, 14:00:18 EDT

The arithmetic step also survives a repeated-root check: it uses the first nonzero derivative, whose order is uniformly bounded away from the unit circle, so it does not require first-order transversality. I have not found a gap in the draft so far. I'm continuing the proof audit and separating the potentially new steps from the Breuillard-Varjú ingredients.

55 Assistant / Progress update / 27 Sep 2026, 14:09:40 EDT

The draft now has a complete candidate argument. The main simplification is to collect variance directly across disjoint digit blocks; this seems to require only approximation at scale $e^{-An \log n}$. I'm checking the scale bookkeeping and arithmetic estimates before treating the argument as reliable.

56 Assistant / Progress update / 27 Sep 2026, 14:10:14 EDT

The literature check found a close precedent for the Taylor step: Dimitrov's argument in Rapaport-Varjú already uses the first nonzero derivative to avoid transversality. I'm correcting the attribution. The potentially new part is the ordered-channel estimate and its direct use in packing variance from digit blocks.

57 Assistant / Progress update / 27 Sep 2026, 14:12:19 EDT

The scale checks still hold: the argument needs very fine low entropy only along a subsequence, while it produces small polynomial values at every sufficiently large degree. Those two different quantifiers let the final contradiction work. The paper is now nine pages, with the borrowed arithmetic arguments explicitly credited.

58 Assistant / Progress update / 27 Sep 2026, 14:17:45 EDT

The most useful idea is to keep variance all the way to the final information bound. Converting it back to entropy introduces another logarithmic loss in the existing approach. I've also checked a stronger consequence of the proposed packing estimate and saved it separately so the main proof stays focused.

59 Assistant / Progress update / 27 Sep 2026, 14:19:59 EDT

The draft and audit are ready, and I'm finishing within the hour limit. No gap emerged in my internal review, but the argument still needs independent mathematical verification. I've marked that status in the paper and identified Lemma 3.4 and Proposition 4.1 as the main steps to scrutinize.

60 Assistant / Reply / 27 Sep 2026, 14:20:39 EDT

I spent **about 48 minutes** and produced a **nine-page candidate proof**, using your usual preamble: [File: varju_gaussian.pdf].

The central idea is to **sum variance directly over disjoint Bernoulli blocks**. An ordered-channel estimate avoids accumulating an error at every scale. This gives the weaker $e^{-An \log n}$ approximation sufficient for the final contradiction.

No gap emerged in my internal audit. **Independent verification and novelty remain unresolved.** The Taylor argument has an established precedent due to Dimitrov, explicitly credited in [Rapaport-Varjú](#).

Editable LaTeX [varju_gaussian.tex] · Detailed audit [audit.md]

For another bounded pass:

/goal Spend at most 60 wall-clock minutes auditing and simplifying the Gaussian-channel proof of Varjú's theorem in this task. Record a deadline and reserve the final five minutes for reporting. Prioritize Lemma 3.4 and Proposition 4.1: seek counterexamples, check conditioning, scale dependence, and block disjointness. Compare potentially new steps with primary literature. Deliver a corrected proof or a precise gap. Preserve my usual AMSart preamble and stop by the deadline.

- Explain the core estimate: Explain the ordered-channel estimate and direct variance packing clearly, with the essential formulas.
- Audit with independent agents: Use independent subagents to audit the candidate proof, especially Lemma 3.4 and Proposition 4.1, and report concrete gaps or verified steps.
- Examine the stronger consequence: Check the stronger scale bound in further-consequence.md and determine whether it follows rigorously and is already known.

61 User / Message / 27 Sep 2026, 14:30:20 EDT

This proof is self-contained, right?

62 Assistant / Progress update / 27 Sep 2026, 14:30:22 EDT

I'll check the draft's dependencies so I can distinguish what it proves from what it cites.

63 Assistant / Reply / 27 Sep 2026, 14:30:37 EDT

Not completely. The draft uses two external results without proving them:

- **Feng-Hu's exact dimensionality theorem**, in Lemma 2.1.
- **Mahler's polynomial root-separation inequality**, in Lemma 5.1.

It also assumes standard tools: de Bruijn's identity, Pinsker's inequality, data processing, Gaussian capacity, and Jensen's formula.

The Breuillard-Varjú entropy-Mahler argument and Dimitrov's Taylor argument **are reproduced**, along with the proposed channel and packing arguments.

So it is **self-contained modulo those inputs**, with exact dimensionality being a substantial dependency. And separately, it remains a candidate proof awaiting independent verification.

64 User / Goal update / 27 Sep 2026, 14:34:21 EDT

now use all that we have understood to try to prove the exact overlaps conjecture in $d=1$. Try to come up with significant new ideas that circumvents the previous difficulties. Maybe you need some new number theoretic input. Don't give up and work hard. Work for 3 hours and stop then if no proof is in sight.

65 Assistant / Progress update / 27 Sep 2026, 14:34:24 EDT

I'll take this as a three-hour research attempt on the full one-dimensional exact overlaps conjecture. I'll record a stopping time, recheck the earlier arguments, and focus on the obstacles caused by arbitrary contractions and translations. Any proposed breakthrough will need a complete proof audit.

66 Assistant / Progress update / 27 Sep 2026, 14:37:49 EDT

Two routes look worth testing. First, the packing estimate may give enough control over which scales have low entropy to work with a growing Mahler bound. Second, the earlier matrix counterexample used a reducible spectrum; I'll check whether using conjugates of a single algebraic number changes that obstruction. Neither route closes the gap yet.

67 Assistant / Progress update / 27 Sep 2026, 14:46:47 EDT

I'll check which parts are proved in the manuscript and which results it still takes as inputs.

68 Assistant / Progress update / 27 Sep 2026, 14:46:59 EDT

Not completely. The Varjú draft proves its proposed Gaussian-channel and packing steps and reproduces the entropy-Mahler argument, but it cites Feng-Hu's exact-dimensionality theorem and classical polynomial root separation. It also uses standard information-theoretic identities.

It is still a candidate proof awaiting independent verification. I'll keep that distinction explicit while continuing the exact-overlaps research.

69 Assistant / Progress update / 27 Sep 2026, 14:47:55 EDT

The information budget appears to extend to unequal contraction ratios: reveal each block after its preceding word, then condition on that block's ratio. Its remaining tail still has the original self-similar law. This could strengthen the analytic part, but the central obstacle remains: turning many extremely close word maps into an exact equality.

70 Assistant / Progress update / 27 Sep 2026, 14:51:16 EDT

The unequal-ratio argument has passed the conditioning checks so far. I'm now testing a different possible simplification: whether sums of independent digits lose less variance than arbitrary distributions at very fine scales. If true, that could improve the packing bound where the arithmetic argument currently needs more control.

71 Assistant / Progress update / 27 Sep 2026, 14:58:46 EDT

One tempting shortcut fails: adding a fair binary digit can increase entropy by $\log 2$ while adding arbitrarily little integrated local variance to a carefully chosen input law. So an improvement would have to use the full structure of the word distribution. I've recorded this counterexample to avoid relying on that false step.

72 Assistant / Progress update / 27 Sep 2026, 15:04:41 EDT

A more concrete idea has emerged on the arithmetic side. Normalizing by the covariance of the full random series seems to give a stronger entropy bound for three digits, uniform in the third translation. I'm checking the algebraic-unit case and the non-Archimedean case separately; this is a candidate partial result, not a proof of the conjecture.

73 Assistant / Reply / 27 Sep 2026, 15:12:15 EDT

Not fully. The Varjú draft proves the proposed channel and packing estimates and reproduces the entropy-Mahler argument, but still invokes:

- Feng-Hu's exact-dimensionality theorem.
- Classical polynomial root separation.
- Standard information-theoretic identities and inequalities.

It is also still a **candidate proof awaiting independent verification**. Calling it completely self-contained or established would overstate its status.

74 Assistant / Progress update / 27 Sep 2026, 15:12:21 EDT

I'm continuing the three-hour attempt. The covariance argument is the strongest new lead so far. I'll first try to break its proof, then investigate whether it can detect enough information to get past the partial three-map bound.

75 Assistant / Progress update / 27 Sep 2026, 15:16:25 EDT

The covariance proof has survived the main algebraic and information-theoretic checks so far. Its limitation is now precise: it detects a nontrivial partition of the three digits, but the full conjecture needs more than that. I'm testing whether the information left inside the unresolved pair can be recovered at later scales.

76 Assistant / Progress update / 27 Sep 2026, 15:22:33 EDT

I found a possible way around one earlier obstacle: decode polynomial coefficients successively, using a variance bound for the unknown tail. This appears to remove the dependence on the polynomial degree from the earlier entropy estimate. I'm checking that the decoding errors still stay controlled after conditioning on previously decoded coefficients.

77 Assistant / Progress update / 27 Sep 2026, 15:27:25 EDT

I'll check which ingredients the draft proves and which it cites, so I can answer precisely.

78 Assistant / Progress update / 27 Sep 2026, 15:27:49 EDT

Not completely. The Varjú draft proves its proposed channel and packing estimates and reproduces the entropy-Mahler argument, but it cites Feng-Hu's exact dimensionality theorem and classical root separation. It also uses standard information-theoretic identities. It is still a candidate proof awaiting independent verification; the ongoing work has not proved the full exact overlaps conjecture.

79 Assistant / Progress update / 27 Sep 2026, 15:28:41 EDT

The sequential decoding estimate has passed the matrix and covariance checks I've run so far. Its error depends on the coefficient energy, with no separate dependence on polynomial degree. That is useful for sparse relations, but it still falls short of the uniform arithmetic estimate needed for the full conjecture.

80 Assistant / Reply / 27 Sep 2026, 15:29:20 EDT

Not fully. The Varjú draft includes proofs of the proposed channel and packing estimates and reproduces the entropy-Mahler argument, but still relies on:

- Feng-Hu's exact dimensionality theorem.
- Classical polynomial root separation.
- Standard information-theoretic identities.

It is also a **candidate proof awaiting independent verification**. The full exact overlaps conjecture remains unproved.

81 Assistant / Progress update / 27 Sep 2026, 15:29:24 EDT

I'm continuing with the missing arithmetic step: whether persistent entropy loss forces a relation of bounded complexity. The new decoding estimate handles sparse relations, so I'll test whether entropy loss itself can produce that sparsity.

82 Assistant / Progress update / 27 Sep 2026, 15:33:48 EDT

The formal power-series problem has a useful entropy gap: a simple conditioning argument gives a lower bound of about 0.8864 for every three-digit relation curve except the two curves where digits coincide. Transferring that bound uniformly to algebraic parameters is still the difficult step; the current estimate does not do it.

83 Assistant / Progress update / 27 Sep 2026, 15:44:55 EDT

I've isolated a precise arithmetic constraint: two independent relations of coefficient norms K_1, K_2 force $M(\eta) \leq K_1 K_2$, regardless of their degrees. The unresolved step is to show that persistent entropy loss supplies a relation with sufficiently small norm. Large collision classes alone do not justify that inference.

84 Assistant / Progress update / 27 Sep 2026, 15:47:34 EDT

I found a sharper estimate that may substantially simplify the analytic argument. For two ordered conditional inputs, the coarse Gaussian channel loses information in proportion to the same conditional-mean variance detected by the fine channel. This appears to remove the additive error per block - and the growing padding it required. I'm checking the bound carefully before using it.

85 Assistant / Progress update / 27 Sep 2026, 15:58:10 EDT

The refined channel bound passes the checks so far. For unequal contraction ratios, the packing argument also appears to need control only of blocks that cross between family ranges. This removes the dependence on the total number of candidate blocks. I'm now testing a separate product-measure inequality that, if proved in the needed local form, could bypass the arithmetic step for homogeneous systems.

86 Assistant / Reply / 27 Sep 2026, 16:01:02 EDT

The full conjecture remains unproved. I've derived a sharper channel estimate that removes the additive error per block and simplifies packing for unequal contractions.

This opens a possible route around number theory, but it requires an **unproved localized product-measure inequality**. The precise reduction and its limitations are in the research note [analytic-reduction.md]. About 93 minutes remain in the requested session.

87 Assistant / Progress update / 27 Sep 2026, 16:01:07 EDT

I'll test the remaining product-measure inequality more aggressively. The key question is whether linear entropy hidden below a coarse scale forces linear integrated variance, even when the sum has complicated near-collisions. A proof could bypass the arithmetic step for homogeneous systems; a counterexample would show where this approach fails.

88 Assistant / Progress update / 27 Sep 2026, 16:10:12 EDT

I'll check which results the draft still cites, so I can distinguish a complete argument from a self-contained one.

89 Assistant / Progress update / 27 Sep 2026, 16:10:39 EDT

Not fully. The Varjú draft still uses Feng-Hu's exact-dimensionality theorem, classical polynomial root separation, and standard information identities. It includes proofs of the main analytic and arithmetic steps, but remains a candidate argument awaiting independent verification. The broader exact-overlaps conjecture is still unproved; I'm continuing the authorized research session.

90 Assistant / Progress update / 27 Sep 2026, 16:11:51 EDT

The strongest remaining analytic route needs an entropy-to-variance inequality for sums of independent digits. The numerical tests have found no counterexample, but they do not prove it. A recent paper gives a useful entropy comparison for Bernoulli sums; I'm checking whether its coupling argument can control the variance quantity needed here.

91 Assistant / Progress update / 27 Sep 2026, 16:18:38 EDT

I found a precise extension of the coupling argument: it bounds quantized entropy using the mass of the most crowded interval. It does not yet give the variance bound. I'm now testing whether entropy lost when adding two independent copies offers a better way to connect the two.

92 Assistant / Progress update / 27 Sep 2026, 16:25:01 EDT

There is a useful exact lemma here: for every weighted sum S of independent fair bits,

$$H(S + S') \leq \frac{3}{2}H(S),$$

where S' is an independent copy. A short XOR argument proves it. The unresolved step is to bound the entropy lost in addition by integrated local variance, with control of the relevant scales.

93 Assistant / Reply / 27 Sep 2026, 16:30:53 EDT

Not completely. The draft still invokes:

- Feng-Hu's exact-dimensionality theorem;
- Mahler's root-separation theorem;
- standard entropy and Gaussian information identities.

It includes proofs of the entropy-Mahler and Taylor-exclusion steps. The accurate description is **a candidate proof with external inputs, awaiting independent verification**. The full exact-overlaps conjecture remains unproved.

94 Assistant / Progress update / 27 Sep 2026, 16:31:01 EDT

I'm continuing with the entropy-of-addition route. I'll try to prove the remaining universal inequality, while testing it against distributions designed to concentrate entropy in rare branches. A counterexample would be just as useful for deciding where to go next.

95 Assistant / Progress update / 27 Sep 2026, 16:33:02 EDT

The scale issue is now more precise: a global inequality for independent, possibly different inputs would imply the localized estimate by conditioning on intervals. That reduction is proved in the notes. The remaining inequality has survived the tests so far, but a Gaussian-shaped example already ruled out one guessed sharp constant.

96 Assistant / Progress update / 27 Sep 2026, 16:42:42 EDT

The full conjecture is still unresolved. I have extended the covariance lemma to arbitrary fixed digit probabilities and arbitrary real translations; it gives a uniform lower entropy bound, but not the full bound needed. The main analytic route still hinges on the unproved entropy-of-addition inequality. About fifty minutes remain.

97 Assistant / Progress update / 27 Sep 2026, 16:52:16 EDT

I'll check the draft's dependencies so I can answer precisely.

98 Assistant / Progress update / 27 Sep 2026, 16:52:37 EDT

Not completely. The Varjú draft quotes **Feng-Hu exact dimensionality** and **Mahler's root-separation inequality**, and uses standard information-theoretic identities. It includes the entropy-Mahler and Taylor-exclusion arguments. It is still a research draft requiring independent verification; the further attempt at the full exact overlaps conjecture remains incomplete.

99 Assistant / Progress update / 27 Sep 2026, 16:54:06 EDT

The new calculation controls the total information lost through grid boundaries across all scales, without paying a separate error at each scale. That removes one obstacle, but a local entropy-versus-dispersion inequality is still missing.

100 Assistant / Progress update / 27 Sep 2026, 16:57:22 EDT

I now have a candidate argument for the missing addition inequality, using random Poisson cuts. I'm checking it carefully: if it holds, it would advance the homogeneous case substantially, but it would not yet settle arbitrary contraction ratios.

101 Assistant / Progress update / 27 Sep 2026, 17:00:33 EDT

There is also a possible way to handle unequal contraction ratios: group letters into fixed large blocks and condition on each block's contraction. This restores independent sums while losing little entropy per letter. I'm auditing the full chain now; I am not yet claiming a proof of the conjecture.

102 Assistant / Progress update / 27 Sep 2026, 17:03:16 EDT

The finite checks passed for the new entropy estimate and the Poisson-partition identities. I also have a candidate connection to the block-packing argument. I'm writing the complete argument separately, with its proof status and external dependencies stated explicitly.

103 Assistant / Progress update / 27 Sep 2026, 17:03:23 EDT

I'm using the PDF skill to prepare a reviewable draft with your usual preamble, then I'll check the rendered pages.

104 Assistant / Progress update / 27 Sep 2026, 17:09:23 EDT

The complete candidate is now written. It uses h_μ , allows overlaps, and covers unequal signed contraction ratios. This version quotes Feng-Hu exact dimensionality and standard Gaussian information identities. I'm checking the full proof and the typeset draft before treating any of it as settled.

105 Assistant / Progress update / 27 Sep 2026, 17:12:05 EDT

The first audit of the complete chain has found no gap. I checked the entropy cost of conditioning on block contractions, the order of parameter choices, and the loss from blocks crossing between selected ranges. This remains an internally checked candidate, not an independently validated resolution.

106 Assistant / Progress update / 27 Sep 2026, 17:16:48 EDT

I've added proofs of the standard Gaussian estimates and the entropy comparisons in an appendix. The new draft now quotes just one substantial theorem: Feng-Hu exact dimensionality. I'm continuing the mathematical audit, particularly the two places where conditioning must preserve independence.

107 Assistant / Reply / 27 Sep 2026, 17:20:23 EDT

Not strictly. The earlier Varjú draft quotes Feng-Hu exact dimensionality and Mahler root separation.

The new draft includes the Gaussian preliminaries and leaves **Feng-Hu exact dimensionality** as its substantial external input. It now contains a complete **candidate** argument for the $d = 1$, h_μ formula with overlaps. No gap has emerged internally, but it is **not independently verified**.

The eleven-page manuscript is [File: poisson_exact_overlaps.pdf]. The research goal remains active for the final audit.

- Audit the addition inequality: Independently audit the Poisson addition inequality, looking for counterexamples or invalid conditioning.
- Audit the packing argument: Independently audit the predictable block packing and its connection to the dimension formula.
- Explain the new idea: Explain the central Poisson-partition idea and why it might remove the arithmetic obstruction.

108 Assistant / Progress update / 27 Sep 2026, 17:20:30 EDT

I'll use the remaining time to reconstruct the two central arguments from scratch: the Poisson entropy inequality and the reduction for unequal contractions. The previous turn produced a complete draft; this pass is aimed at finding a flaw in it.

109 Assistant / Progress update / 27 Sep 2026, 17:21:16 EDT

The fresh check agrees with the Poisson entropy calculation. I also found a more elementary proof of the dispersion comparison: it reduces to one explicit integral identity. I'm replacing that step to make the argument easier to verify.

110 Assistant / Progress update / 27 Sep 2026, 17:24:23 EDT

The conditioning check also passes for unequal signed ratios with genuine exact overlaps; direct word enumeration agrees with the conditional product law. The boundary-block check passes too. These are supporting checks of finite identities - the dimension conclusion still rests on the written argument and needs independent scrutiny.

111 Assistant / Progress update / 27 Sep 2026, 17:26:46 EDT

The simplified step is the identity

$$\int_0^A \frac{s(2d-s)}{(d+s)^4} ds = \frac{A^2}{(d+A)^3} \geq 0.$$

It gives the required dispersion-to-variance comparison by weighting with a decreasing function. This makes one of the new argument's central estimates directly checkable.

112 Assistant / Progress update / 27 Sep 2026, 17:30:08 EDT

I've found a possible substantial simplification in the final check: the Poisson inequality may make the entire block-packing section unnecessary. A single long prefix appears to have only $o(n)$ information left to reveal below its natural scale, while the new variance bound would force a gain of order n . I'm checking that direct contradiction now.

113 Assistant / Progress update / 27 Sep 2026, 17:31:49 EDT

The direct prefix argument checks out internally. It also removes the need for the sharper coarse Gaussian estimate: with only one prefix, a fixed error is harmless beside $o(n)$. I'm shortening the manuscript accordingly. A complete proof candidate is now in sight, so this final simplification uses the continuation you allowed at the three-hour mark.

114 Assistant / Progress update / 27 Sep 2026, 17:37:09 EDT

I'll check the current manuscript's dependencies so I can distinguish what it proves in full from what it cites.

115 Assistant / Progress update / 27 Sep 2026, 17:37:50 EDT

The current draft is not fully self-contained: it quotes Feng-Hu's exact-dimensionality theorem. It supplies proofs of the finite-input estimates and Gaussian tools it uses. Those new arguments remain a candidate proof requiring independent verification.

116 Assistant / Reply / 27 Sep 2026, 17:37:58 EDT

Not completely. The current draft quotes **Feng-Hu's exact-dimensionality theorem**. It proves the addition inequalities, variance estimates, and Gaussian preliminaries, starting from standard entropy identities and measure theory.

Thus it is self-contained **apart from exact dimensionality**. Making it fully self-contained would require proving the needed case of Feng-Hu.

Separately, the new argument remains an **unverified proof candidate**, not an established resolution of the conjecture.

117 Assistant / Progress update / 27 Sep 2026, 17:38:05 EDT

I'll make a fresh adversarial check of the two central claims: the bound on information lost under addition, and the conversion of that bound into a dimension estimate. The last turn clarified dependencies but made no new mathematical progress.

118 Assistant / Progress update / 27 Sep 2026, 17:39:52 EDT

The latest checks have not exposed an error. I found a cleaner way to organize the final step: an explicit finite inequality can connect $H(R_n, B_n)$ directly to the fine-scale variance. This will make the conditioning costs and order of limits easier to check.

119 User / Message / 27 Sep 2026, 17:39:53 EDT

Can you show me the pdf you currently have

120 Assistant / Progress update / 27 Sep 2026, 17:39:56 EDT

I'll open the current ten-page draft. It is the proof candidate with the direct prefix-information argument.

121 Assistant / Reply / 27 Sep 2026, 17:40:03 EDT

Here is the current ten-page draft, **Entropy of addition and self-similar measures on the line**: [File: poisson_exact_overlaps.pdf]

It remains an unverified proof candidate.

122 Assistant / Progress update / 27 Sep 2026, 17:40:19 EDT

I'm now writing out the finite entropy comparison that closes the proposed argument. The PDF you just saw remains unchanged while I check this reformulation.

123 User / Message / 27 Sep 2026, 17:43:22 EDT

Ok stop the goal for now and I update the goal in a moment. Give me the pdf

124 Assistant / Progress update / 27 Sep 2026, 17:43:25 EDT

I'll pause the goal and open the latest PDF, which includes the finite entropy comparison.

125 Assistant / Reply / 27 Sep 2026, 17:43:32 EDT

The goal is **paused**.

Here is the latest ten-page PDF, including the simplified final argument: [File: poisson_exact_overlaps.pdf]

It remains an unverified proof candidate.

126 User / Goal update / 27 Sep 2026, 17:45:34 EDT

Rewrite the paper poisson_exact_overlaps.pdf so that it is publication ready and professional. Write it stylewise like this paper <https://arxiv.org/abs/2609.26771>. Do the following:

- 1) Quote all the relevant literature before stating the main theorem. This includes Hochman, Varju, Varju-Rapaprot, Rapaport,
- 2) Explain the main proof ideas
- 3) Check the proof extremely precisely. You are making a big claim here.

127 Assistant / Progress update / 27 Sep 2026, 17:45:38 EDT

I'll read the reference paper, verify the relevant literature, and audit the argument before rewriting it. I'll preserve your preamble and use the PDF skill to check the finished layout. If the audit finds a gap, I'll make that explicit in the manuscript.

128 User / Message / 27 Sep 2026, 17:46:07 EDT

And also have the same level of detail as in the paper I have sent you

129 Assistant / Progress update / 27 Sep 2026, 17:46:11 EDT

I'll match that level of detail: a literature-led introduction, a proof outline with the key estimates, and full explanations of conditioning, averaging, and limiting steps. The draft currently compresses several of those steps into a sentence; I'll expand them.

130 Assistant / Progress update / 27 Sep 2026, 17:47:34 EDT

The literature review needs to distinguish several different cases: algebraic contractions, rational or algebraic translations, and transcendental Bernoulli convolutions. I've checked those statements against the papers. In the proof audit, I'm expanding the Poisson differentiation, the Gaussian grid averaging, and the order of limits - the steps carrying most of the new claim.

131 Assistant / Progress update / 27 Sep 2026, 17:51:42 EDT

I found a missing hypothesis in the ordered-detection lemma: the background must be independent of the pair consisting of the input and its binary label. Every application here satisfies that condition because the label is a function of the input. I'm stating it explicitly and separating out the first-letter information lemma so its use later is clear.

132 Assistant / Progress update / 27 Sep 2026, 17:57:01 EDT

The rewrite now puts all 14 cited works before the main theorem and gives a detailed proof outline. I've also expanded the delicate arguments and corrected the lemma's independence hypothesis. I'm checking the final pages and recording the audit separately; this is still my own review, not independent verification.

133 Assistant / Progress update / 27 Sep 2026, 18:01:13 EDT

The manuscript is now 16 pages, with the conditioning arguments and limiting steps written out in detail. The audit caught and corrected a missing joint-independence assumption in the detection lemma. All pages render cleanly; I'll deliver it as a candidate proof that still needs independent mathematical review.

134 Assistant / Reply / 27 Sep 2026, 18:02:15 EDT

Expanded to **16 pages**, with detailed proofs, a substantial proof outline, and all 14 references discussed before the main theorem. Your preamble is preserved.

Here is the revised manuscript: [File: poisson_exact_overlaps.pdf].

The audit corrected a missing joint-independence hypothesis. **The proposed theorem still requires independent verification.** The detailed audit [professional-rewrite-audit.md] records the checks and limitations.

The rewrite goal is complete. Recorded goal time: approximately 3 hours 26 minutes.

- Prepare a referee packet: Prepare a concise referee packet identifying the new claims and the steps most needing independent verification.
- Isolate the addition inequality: Extract the Poisson entropy-of-addition inequality into a standalone note with complete hypotheses and proof.
- Specialize to Bernoulli convolutions: Write a streamlined version of the candidate argument for transcendental Bernoulli convolutions, clearly preserving its verification status.

135 User / Message / 27 Sep 2026, 18:40:38 EDT

How easily do you think does this generalize to $\mathbb{R}^d$? Don't start yet too heavily, just tell me what you think the main bottlenecks in this case are

136 Assistant / Progress update / 27 Sep 2026, 18:40:40 EDT

I'll look at which steps use the order on the line and compare the higher-dimensional results, then give you a preliminary assessment.

137 Assistant / Reply / 27 Sep 2026, 18:41:38 EDT

My preliminary view is that **the overall strategy may extend, but the general case needs substantial new ideas**. This assessment is conditional on the present one-dimensional candidate surviving independent checking.

The Gaussian entropy identities, first-letter information argument, and prefix-saturation argument should transfer fairly directly: isotropic Gaussian noise behaves well under similarities in every dimension. I see four main bottlenecks.

1. **The statement needs a geometric hypothesis.** In higher dimensions, a measure can already have full dimension along one subspace while remaining deficient in another. Thus the unrestricted formula

$$\dim \nu = \min\{d, h_\mu/\chi\}$$

cannot simply be carried over. Irreducibility of the orthogonal parts, together with no common fixed point, is the natural initial setting. This is the setting of Kittle-Kogler; Hochman's higher-dimensional theorem explicitly identifies the obstruction from invariant subspaces. [Kittle-Kogler](#), [Hochman](#).

2. **Detecting variance becomes directional. This is the main analytic difficulty.** On the line, dimension deficiency produces roughness in the only available direction. In $\mathbb{R}^d$, the same calculation controls the **trace** of Fisher information; it does not immediately give roughness in the direction where the input has variance. A product of Lebesgue measure and a Cantor measure illustrates the problem: one direction is smooth, another is rough.

Moreover, our ordered-detection lemma uses a special one-dimensional fact: monotone coupling makes the expected displacement equal to the difference of means. Vector displacements can cancel. We would need a replacement that relates the input's covariance to the background's directional roughness, probably using irreducibility to spread directions. Projecting onto lines alone is insufficient: projections can have full dimension even when the original measure does not.

3. **The Poisson-cut argument needs new geometry.** Random hyperplanes are a natural replacement for cuts at points. There is a promising foothold: conditioning on the full vector sum gives at least as much information as conditioning on its projection, so the one-cut entropy bound has a projected version. But our comparison between integrated dispersion and local variance uses the precise geometry of intervals. Replacing that with an estimate for higher-dimensional cells requires a new argument.

4. **Conditioning on rotations can cost linear entropy.** The current proof crucially uses

$$H(R_N) = O(\log N),$$

because scalar products are determined by letter counts. For matrix products L_N , noncommuting rotations can instead give $H(L_N)$ proportional to N . Conditioning on all block linear parts would then discard a substantial part of h_μ . Finite rotation groups avoid this problem; commuting rotations also have inexpensive conditioning. This makes the planar case more approachable than general noncommuting systems in higher dimensions.

My first test case would be $d = 2$ with a finite irreducible rotation group. It removes the rotation-entropy obstacle while retaining the central geometric difficulty. I would first investigate a replacement for the localized variance-detection lemma; that should tell us whether the present proof offers a short higher-dimensional route.

138 User / Message / 27 Sep 2026, 18:44:27 EDT

Ok. Now let's focus on the 1d case. Looks very interesting. Try hard to find any significant simplifications that make the proof easier (but ideally still self-contained)

139 Assistant / Progress update / 27 Sep 2026, 18:44:30 EDT

I'll look for steps we can remove or combine, especially in the Poisson argument and the passage from entropy to variance. I'll also check whether each simplification preserves the hypotheses and constants the final proof needs.

140 Assistant / Progress update / 27 Sep 2026, 18:45:50 EDT

There's a promising way to use the same Poisson partitions on both sides of the proof. They may also give the Gaussian variance-detection estimate, which would remove the separate dyadic partitions, the second variance profile, and the comparison between them. I'm checking the identity that would make this work.

141 Assistant / Progress update / 27 Sep 2026, 18:48:37 EDT

I've found two more simplifications: the product estimate can use Gaussian noise throughout, eliminating the separate uniform-noise entropy; and prefix saturation follows from a short chain-rule bound. The Poisson cut identity also checks out algebraically. I'm writing out its stopping-partition argument carefully before putting it into the revised draft.

142 Assistant / Progress update / 27 Sep 2026, 18:53:24 EDT

The revised argument now uses one variance energy and one family of Poisson partitions. The key identity says that the expected sum of squared child-mean differences equals that energy plus a nonnegative boundary term. I've checked the algebra exactly on 224 finite examples; the proof is written out as well.

The rewrite also removes the uniform-noise localization lemma and shortens prefix saturation to a chain-rule sandwich. It still uses Feng-Hu exact dimensionality, so it is not strictly self-contained.

143 Assistant / Progress update / 27 Sep 2026, 18:57:11 EDT

The revised draft is 15 pages and retains the detailed proofs. Its main simplification is structural: addition and detection now share one Poisson calculation, Gaussian entropy is used throughout, and the block trick appears only at the end. I'm finishing the consistency checks and keeping the previous version for comparison.

144 Assistant / Reply / 27 Sep 2026, 18:58:19 EDT

I found **four substantial simplifications** and incorporated them into the manuscript.

1. **One Poisson argument for both halves of the proof.** The expected sum of normalized between-child variances equals the local variance energy plus a nonnegative boundary term. This removes the separate dyadic partitions, auxiliary variance profile, and comparison between profiles. The dispersion estimate also gets a simpler proof using exponential waiting times.

2. **Gaussian entropy throughout.** Applying the addition inequality inside ordinary interval cells gives the finite-product estimate directly. The separate uniform-noise localization lemma disappears.

3. **A much shorter prefix-saturation proof.** The chain rule and monotonicity give

$$\mathbb{E}I_1(t/|R_n|) \leq \frac{1}{n}I(W_n; X + tZ) \leq \alpha\chi.$$

Setting $t = e^{-bn}$, $b > \chi$, finishes the argument by bounded convergence. No subtraction of two large entropy asymptotics is needed.

4. **The block trick moves entirely to the end.** First obtain

$$h_\mu \leq H(R_1) + \alpha\chi.$$

Applying this to the N -fold system removes the ratio cost because $H(R_N) = o(N)$. This substantially simplifies the parameters in the intermediate estimates.

The revised **15-page manuscript**, retaining your preamble and detailed proofs, is here: [File: poisson_exact_overlaps.pdf].

The derivations and checks [poisson-simplifications.md] are recorded separately. These changes add no external theorem, but **Feng-Hu exact dimensionality remains an input**, and the main claim still needs independent verification.

- Explain the key identity: Explain the new Poisson splitting identity from first principles, including a two-point example.
- Attack the detection estimate: Try to falsify the new Poisson variance-detection estimate and identify any hidden assumptions.
- Remove the remaining external input: Investigate a self-contained replacement for the quoted exact-dimensionality theorem in this one-dimensional setting.

145 User / Message / 27 Sep 2026, 19:08:40 EDT

A few things:

- 1) Instead of χ write χ_{μ} throughout the paper
- 2) use $\frac{h_{\mu}}{\chi_{\mu}}$ everywhere in the paper instead of h_{μ}/χ_{μ}
- 3) Write the proofs as elegantly as you can without omitting details
- 4) Make the introduction shorter. You don't need to explain the proofs of the previous results, just make the statements shorter
- 5) Change the title to "The generalized exact overlaps conjecture on the dimension of self-similar measures on the line" and the running title "The generalized exact overlaps conjecture"

146 Assistant / Progress update / 27 Sep 2026, 19:08:51 EDT

I'll update the notation and titles, shorten the literature discussion, and polish the proofs while preserving the details. Then I'll rebuild and check the PDF.

147 Assistant / Progress update / 27 Sep 2026, 19:11:58 EDT

The notation and titles are updated. The introduction is about 39% shorter, with the literature statements and references retained. I've polished the proofs and kept all 14 proof blocks; I'm now checking the equations and page layout.

148 Assistant / Progress update / 27 Sep 2026, 19:14:09 EDT

The page review is complete. I'm making two small typography fixes, then rebuilding the final PDF and checking the notation once more.

149 Assistant / Reply / 27 Sep 2026, 19:15:08 EDT

All five changes are applied: consistent χ_μ and $\frac{h_\mu}{\chi_\mu}$, polished proofs retaining the details, a shorter introduction, and your exact title and running title. Your preamble is unchanged.

The revised 15-page PDF compiles cleanly and has been visually checked: [File: poisson_exact_overlaps.pdf]

150 User / Message / 27 Sep 2026, 19:31:24 EDT

Where is this Variance energy coming from? Is it completely new?

151 Assistant / Progress update / 27 Sep 2026, 19:31:27 EDT

I'll trace its role in the argument and check the related literature so I can distinguish what is standard from what might be new.

152 Assistant / Progress update / 27 Sep 2026, 19:32:06 EDT

There is a useful interpretation: the energy is the squared error remaining after observing the variable with uniform noise, integrated over logarithmic scales. That makes its normalization natural. The part whose novelty needs checking is the Poisson-partition argument connecting it to entropy.

153 Assistant / Reply / 27 Sep 2026, 19:32:41 EDT

I would not claim that it is completely new. It is a natural way of integrating local conditional variance over scales. The potentially new part is how the draft connects this quantity to entropy through Poisson partitions; I have not established priority for those estimates.

In our earlier drafts, the starting point was the normalized local variance

$$V_\rho(r) = \frac{1}{r^3} \int_{\mathbb{R}} \rho([a, a+r)) \operatorname{Var}(\rho \mid [a, a+r)) da.$$

Its normalization has a simple explanation: averaging over the origin of a grid contributes $1/r$, and dividing the variance by the squared cell size contributes $1/r^2$. Thus $V_\rho(r)$ measures the uncertainty remaining inside a typical cell, relative to its size.

The energy simply adds these contributions over logarithmic scales:

$$\mathcal{E}_R(\rho) = \int_0^R V_\rho(r) \frac{dr}{r}.$$

This explains the r^{-4} in the paper. It also explains why the full energy is unchanged by rescaling the measure.

There is an especially useful interpretation that the current exposition does not spell out. Let $Y \sim \rho$ and let U be independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$. Then, **exactly**,

$$\mathcal{E}_R(\rho) = \int_0^R \frac{\mathbb{E} \operatorname{Var}(Y \mid Y + rU)}{r^2} \frac{dr}{r}.$$

Indeed, conditional on $Y + rU = z$, the law of Y is precisely ρ restricted to the interval $[z - r/2, z + r/2]$; the observation density is that interval's mass divided by r . Integrating gives the identity.

So the energy is an **integrated, normalized minimum mean-square estimation error with uniform noise**. This places it near the classical connection between information and estimation, notably the Gaussian I-MMSE identity of [Guo-Shamai-Verdú](#). Their identity concerns Gaussian noise; it does not directly establish our uniform-noise estimates.

It is also closely related in spirit to your work with Kittle: normalized conditional variance at a scale, followed by accumulation across scales. Your trace functional optimizes over suitable conditional decompositions, whereas this functional uses an explicit average over intervals. They are different quantities within the same general approach. See [Kittle-Kogler, §2.1](#).

The reason this particular energy became useful here is its exact Poisson representation:

$$6\mathcal{E}(\rho) = \int_0^\infty t \mathbb{E} \text{Var}(Y \mid Q_t, Q_t(Y)) dt,$$

where Q_t is an independent Poisson partition of intensity t . As cuts arrive, entropy decomposes by the chain rule, while variance decomposes by the law of total variance. The draft's proposed contribution is to make these two decompositions control the **same energy**, both in the addition inequality and in the detection argument.

I would therefore credit the established variance-summation and information-estimation background explicitly, and treat the specific Poisson identities and entropy bounds as candidates for novelty pending a more thorough literature comparison and independent checking.

154 User / Message / 27 Sep 2026, 19:47:51 EDT

What is $\mathcal{D}(X)$ as used in Lemma 2.2?

155 Assistant / Reply / 27 Sep 2026, 19:47:58 EDT

$$\mathcal{D}(X) = \mathbb{E}|X - X'|,$$

where X' is an independent copy of X . Thus it is the **mean absolute distance between two independent samples**.

For a finite law $\mathbb{P}(X = x_i) = p_i$,

$$\mathcal{D}(X) = \sum_{i,j} p_i p_j |x_i - x_j|.$$

156 User / Message / 27 Sep 2026, 19:52:15 EDT

Is it maybe more natural to look at the normalized local variance?

157 Assistant / Reply / 27 Sep 2026, 19:52:44 EDT

Yes. **I think normalized local variance is the more natural primary quantity**, with the energy introduced afterwards as its integral over scales.

For $X \sim \rho$, define

$$V_\rho(r) = \frac{1}{r^3} \int_{\mathbb{R}} \rho([a, a+r)) \text{Var}(\rho \mid [a, a+r)) da.$$

Equivalently, choose a uniformly shifted grid of mesh r , reveal the cell containing X , and take the expected conditional variance divided by r^2 . Thus $V_\rho(r)$ directly measures the uncertainty remaining at resolution r .

The energy is then simply

$$\mathcal{E}_R(\rho) = \int_0^R V_\rho(r) \frac{dr}{r}.$$

This presentation makes the argument easier to interpret: establish estimates for uncertainty at individual scales, then add them over logarithmic scales.

There is one distinction concerning Lemma 2.2. Its auxiliary quantity $\mathcal{D}(X)$ cannot simply be replaced by variance at one fixed scale. If X is supported in an interval of length r , then

$$2 \frac{\text{Var}(X)}{r^2} \leq \frac{\mathcal{D}(X)}{r} \leq \sqrt{2 \frac{\text{Var}(X)}{r^2}}.$$

There is generally no upper bound linear in the normalized variance: two equally weighted atoms separated by εr have normalized dispersion $\varepsilon/2$, but normalized variance $\varepsilon^2/4$.

That is precisely why the integration over scales matters. The Poisson argument converts the accumulated dispersion into accumulated normalized variance; it does not make that conversion separately at every scale.

I would therefore organize the exposition around $V_\rho(r)$, retain $\mathcal{E}_R$ as convenient notation for its integral, and introduce $\mathcal{D}$ only within the auxiliary entropy estimate.

158 User / Message / 27 Sep 2026, 19:53:35 EDT

Ok write it more like the earlier draft then. It is a bit a lot to swallow do directly go to $\mathcal{E}_R(\rho)$. But still try to be elegant, but don't hide ideas that makes it easier to understand

159 Assistant / Progress update / 27 Sep 2026, 19:53:38 EDT

I'll introduce normalized local variance first, explain its meaning at a single scale, and then build up to the energy integral. I'll also make the role of dispersion clearer before it enters the proof.

160 Assistant / Progress update / 27 Sep 2026, 19:54:25 EDT

The revision will follow the quantities in the order they arise: variance inside a grid cell, normalization by the cell size, and then integration over scales. I'll keep the Poisson argument, with clearer explanations of how each cut reveals information and variance.

161 Assistant / Progress update / 27 Sep 2026, 19:56:51 EDT

The new exposition starts with a grid-cell interpretation and a two-atom example before introducing $\mathcal{E}_R$. It also explains what $\mathcal{D}$ measures and why dispersion is converted to variance only after summing over scales. The added explanation brings the paper to 16 pages; I'm checking the layout and cross-references now.

162 Assistant / Reply / 27 Sep 2026, 19:58:27 EDT

Revised: normalized local variance now comes first, with its grid interpretation and a two-atom example. The energy integral follows naturally, and the roles of dispersion and binary cuts are explained before use. Proof details and your preamble are preserved.

The checked 16-page PDF is here: [File: poisson_exact_overlaps.pdf]

- Explain the Poisson identity: Walk me through the Poisson variance identity step by step.
- Audit the addition inequality: Carefully audit the entropy-of-addition inequality for gaps.

- Compare with Kittle-Kogler: Compare the normalized local variance precisely with the trace functional in Kittle-Kogler.

163 User / Message / 28 Sep 2026, 05:19:18 EDT

Give me the latex so that I can upload it on prism. Put the different sections into different files so that it is easy to work with

164 Assistant / Progress update / 28 Sep 2026, 05:19:28 EDT

I'll package the LaTeX with a main file, your preamble, and a separate file for each section, then check that it compiles.

165 Assistant / Reply / 28 Sep 2026, 05:21:01 EDT

Download the LaTeX project ZIP [exact-overlaps-prism.zip].

Each section has its own file, alongside separate files for the appendix, references, abstract, and preamble.

Use `main.tex` as the main document. Your preamble is preserved verbatim, and the project compiles without warnings to the identical 16-page paper.

166 User / Message / 28 Sep 2026, 05:28:26 EDT

It didn't work to compile for some reason. check again and make sure that it compiles

167 Assistant / Progress update / 28 Sep 2026, 05:28:28 EDT

I'll check the ZIP from a fresh extraction and look for anything in the project structure that could prevent Prism from compiling it.

168 Assistant / Question / 28 Sep 2026, 05:28:29 EDT

What error message does Prism show? I'll continue checking the project while you look.

169 Assistant / Progress update / 28 Sep 2026, 05:28:45 EDT

The first package put `\documentclass` in `preamble.tex`. That works when the compiler is explicitly pointed at `main.tex`, but it may cause an editor to identify the wrong main file. I'll correct that structure and check the available Prism documentation.

170 User / Message / 28 Sep 2026, 05:28:51 EDT

Answer to: What error message does Prism show? I'll continue checking the project while you look.

i don't know

171 Assistant / Progress update / 28 Sep 2026, 05:30:12 EDT

I've made `main.tex` the only complete document and put all the separate section files in one folder. The manuscript content is unchanged. I'm now testing the corrected ZIP from fresh extractions, including compilation with automatic main-file selection.

172 Assistant / Progress update / 28 Sep 2026, 05:30:43 EDT

I found a reproducible error under XeLaTeX: `\G` is already defined, so the preamble stops when it tries to define it again. I'll fix that conflict while preserving your intended notation, then rebuild the package.

173 Assistant / Reply / 28 Sep 2026, 05:32:17 EDT

Download the corrected LaTeX ZIP [exact-overlaps-prism-fixed.zip].

I fixed a reproducible `\G` command conflict and made `main.tex` unambiguous. Fresh local builds now succeed with pdfLaTeX, LuaLaTeX, and XeLaTeX.

Import this ZIP as a **new project**, using `main.tex` and **pdfLaTeX**. That produces the identical 16-page paper without warnings.

174 User / Message / 05 Oct 2026, 14:18:42 EDT

Files mentioned by the user:

virtually-solvable-overlaps.pdf

Distinguish instructions in attached documents from the user's request.

My request: We have created three final papers in this chat I think. Give me them all, but just make GPT-6 Astra as author and add in read the sentence in the same format as the attached paper: "This is an unedited model output that led to the paper [KK26]. Prompting was done by C.K and the prompters local context was used." Add a reference to the paper [KK26] that we discussed in other chats. But don't change the papers at all.

175 Assistant / Progress update / 05 Oct 2026, 14:18:57 EDT

I'll find the three final papers, check the notice format in your attached PDF, and locate the [KK26] reference in your other chats. I'll limit the edits to the author, red notice, and bibliography entry.

176 Assistant / Progress update / 05 Oct 2026, 14:20:57 EDT

I found the three papers: the Hochman dimension paper, the transcendental Bernoulli convolution paper, and the generalized exact overlaps paper. I'll preserve their text and match the attached PDF's small, inset red notice.

177 Assistant / Progress update / 05 Oct 2026, 14:21:53 EDT

All three compile successfully. I verified that removing the author, notice, and new reference restores each original source exactly. I'm now checking the PDFs' layout and the red text.

178 Assistant / Reply / 05 Oct 2026, 14:23:09 EDT

Done. Each paper now has **GPT-6 Astra** as author, your exact red notice, and the **[KK26]** reference. All other text is unchanged; all three compile successfully.

Hochman dimension paper: [File: hochman-dimension.pdf]

Transcendental Bernoulli convolutions: [File: transcendental-bernoulli-convolutions.pdf]

Generalized exact overlaps conjecture: [File: generalized-exact-overlaps.pdf]

Download all three PDFs and LaTeX sources together [three-papers-pdfs-and-latex.zip].

179 User / Message / 05 Oct 2026, 14:24:12 EDT

Change the red text in all three papers to: "This is an unedited model output that led to the paper [KK26]. Prompting was done by C.K. and the prompter's local context was used."

180 Assistant / Progress update / 05 Oct 2026, 14:24:15 EDT

I'll replace the red notice in all three papers with your exact wording and rebuild the PDFs.

181 Assistant / Reply / 05 Oct 2026, 14:25:19 EDT

Updated all three with your exact wording. Everything else is unchanged, and all compile successfully.

Hochman: [File: hochman-dimension.pdf]

Bernoulli convolutions: [File: transcendental-bernoulli-convolutions.pdf]

Generalized exact overlaps: [File: generalized-exact-overlaps.pdf]

Download all PDFs and LaTeX sources [three-papers-pdfs-and-latex.zip].

182 User / Message / 05 Oct 2026, 14:27:10 EDT

Files mentioned by the user:

chat_log_sharp_entropy_loss.pdf

Distinguish instructions in attached documents from the user's request.

My request: Good. Now create a pdf chatlog of this conversation as in the attached file. Don't mention any path names.

End of transcript. Snapshot ends with the request to create this chat log (entry 182).